

Paper F-04: Nuclear Dynamics: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

Founder, Vura Labs

Kampala, Uganda

ORCID: 0009-0001-0556-5516

August 2026

Shared Part IV notation and evidence boundaries are defined in PART_IV_METHODS_AND_CLAIM_BOUNDARY.md. This paper states only its local observables, comparison, and failure conditions.

Abstract

This paper develops a falsifiable CDFD/AFL model of Nuclear Dynamics. It maps particle, energy, and reaction fluxes to driving flux Φ , binding, conservation laws, Coulomb barriers, and available states to active constraint C , decay, reaction channels, collective modes, and energy redistribution to responsiveness S , and isotopic composition, excitation, deformation, and reaction history to structural memory M_s . The target outcome is stability, yields, decay pathways, and energy release, tested against cross sections, decay data, spectroscopy, collision experiments, and nuclear models. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

1 Introduction

At the femtometer scale, the atomic nucleus is a highly dense, dynamic droplet of nucleons bound together by the residual strong force. The balance of forces within the nucleus dictates the stability of all matter in the universe. While quantum chromodynamics (QCD) describes the fundamental quark-gluon interactions, describing bulk nuclear phenomena (such as fission and alpha decay) requires a macro-scale thermodynamic approach.

CDFD offers a unified thermodynamic ontology. Whether analyzing a living biological cell or an atomic nucleus, the system must maintain sufficient internal throughput to offset its structural constraints. In the nucleus, this throughput is the continuous exchange of virtual mesons (primarily pions) between nucleons.

2 The Nuclear Adaptive Surface

We model the atomic nucleus as an active surface characterized by the CDFD scaling equation:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

For a nucleus composed of Z protons and N neutrons, the parameters map as follows:

- **Flux (Φ):** The strong force mediated nucleon-nucleon exchange. This flux scales with the total number of adjacent nucleon pairs, roughly proportional to the nuclear volume $A = Z + N$.
- **Constraint (C):** The disruptive Coulomb repulsion between protons, which scales as $Z^2/A^{1/3}$, along with the Pauli exclusion principle which forces nucleons into higher energy states.
- **Responsiveness (S):** The binding energy per nucleon, indicating the depth of the local potential well and the surface tension of the nuclear droplet.
- **Memory (M_s):** Nuclear shell structure. Nuclei with "magic numbers" of protons or neutrons exhibit closed shells, locking the nuclear topology into a highly efficient, rigid state (perfect memory, $M_s \rightarrow 1.0$).

2.1 Nuclear Decay as Adaptive Failure

A stable nucleus maintains $\Psi_s \approx 1$. The cohesive flux of the strong force perfectly regulates the outward pressure of the Coulomb constraint.

As we move to heavy elements (increasing Z), the Coulomb constraint $C \propto Z^2$ grows much faster than the strong force flux $\Phi \propto A$. Consequently, the ratio (Φ/C) begins to plummet. To prevent the system from entering a catastrophic decay regime ($\Psi_s \ll 1$), the nucleus must adapt by increasing its neutron-to-proton ratio, effectively boosting Φ without adding to C .

However, eventually, the surface responsiveness S reaches a saturation limit. The nucleus can no longer bind the excess neutrons effectively due to the Pauli constraint. The system memory M_s locks into a state of chronic overload. The nucleus violently sheds the excess constraint through Alpha decay or spontaneous fission, fracturing the system to restore $\Psi_s \approx 1$ in the resulting daughter nuclei.

3 Measurement Layer

4 Cross-Domain Model Upgrade: Mechanism, Scale, and Tests

4.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Nuclear Dynamics, the proposed drive is particle, energy, and reaction fluxes. The active limitation is binding, conservation laws, Coulomb barriers, and available states. Responsiveness

is carried by decay, reaction channels, collective modes, and energy redistribution, while retained state is represented by isotopic composition, excitation, deformation, and reaction history. The outcome to explain is stability, yields, decay pathways, and energy release.

Table 1: Operational CDFD/AFL map for Paper F-04.

Term	Paper-specific observable meaning	Measurement question
Φ	particle, energy, and reaction fluxes	What rate, load, gradient, or demand enters the focal system?
C	binding, conservation laws, Coulomb barriers, and available states	Which measured bottleneck limits transfer, function, or recovery?
S	decay, reaction channels, collective modes, and energy redistribution	Which process changes effective capacity or reroutes the load?
M_s	isotopic composition, excitation, deformation, and reaction history	Which retained state makes equal present loads produce different futures?
Y	stability, yields, decay pathways, and energy release	Which outcome is predicted out of sample, with uncertainty?

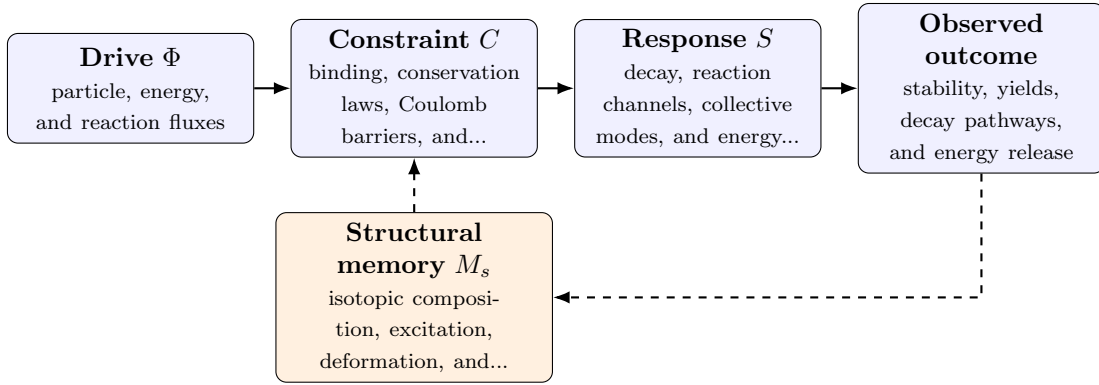


Figure 1: Paper F-04 observable CDFD/AFL architecture for Nuclear Dynamics. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

4.2 Paper-Specific Causal Hypothesis

For this paper, the hypothesized causal sequence is specific. A change in particle, energy, and reaction fluxes arrives faster than decay, reaction channels, collective modes, and energy redistribution can compensate. This raises or redistributes binding, conservation laws, Coulomb barriers, and available states. If the load persists, the retained state encoded by isotopic composition, excitation, deformation, and reaction history changes the next response, so a later event of equal magnitude can produce a different trajectory in stability, yields, decay pathways, and energy release. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

4.3 Cross-Scale Translation Without Mechanistic Erasure

For the cosmic and subatomic systems family, the cross-domain translation must remain subordinate to conservation laws, relativity, quantum mechanics, and established astrophysical or plasma equations. CDFD is useful only if it compresses a regime transition or hysteresis question

without pretending that biological adaptation and physical response are the same mechanism. [3, 2, 1] The required scale declaration for this paper is attoseconds to geological time; nucleons to astrophysical reaction networks. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

4.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper F-04. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure cross sections, decay data, spectroscopy, collision experiments, and nuclear models and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a controlled energy or particle-flux scan across isotopes while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper's stated falsifier is: the translation adds no testable structure beyond established nuclear theory.

4.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from cross sections, decay data, spectroscopy, collision experiments, and nuclear models and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of stability, yields, decay pathways, and energy release. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the cross-domain translation audit.

5 Conclusion

Radioactive decay is not a random violation of stasis; it is the necessary, model-predicted response of an active system failing to maintain adequate throughput against rising constraints. By framing nuclear dynamics as an Adaptive Flux Limited system, we bridge the gap between subatomic quantum mechanics and universal macroscopic thermodynamics, revealing that the atomic nucleus breathes and regulates its structural topology exactly like an organic cell.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

6 Reference Layer

References

- [1] Per Bak, Chao Tang, and Kurt Wiesenfeld. Self-organized criticality: An explanation of the $1/f$ noise. *Physical Review Letters*, 59(4):381–384, 1987. DOI: <https://doi.org/10.1103/PhysRevLett.59.381>.
- [2] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948. DOI: <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>.
- [3] A. M. Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London. Series B*, 237(641):37–72, 1952. DOI: <https://doi.org/10.1098/rstb.1952.0012>.